NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Electronics and Communication Engineering
MID-SEMESTER EXAMINATION
Session Jul-Dec 2024

B.Tech.: Semester-I
Course Name: Digital Design
Maximum Time: 2 hours **Course Code: EC14102**
Max. Marks: 30

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome), and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1.	Answer all the questions (each 1 Marks)	10	CO1, CO2	U, A, R, E,

a) In a 4-variable Karnaugh map, a 3-variable POS term is produced by _____ cell group of 0's.

b) Prove that $\overline{(\overline{AB} + \bar{A} + AB)} = 0$.

c) Convert the following numbers with the indicated bases to decimal:
(i) $(4310)_6$ (ii) $(198)_{12}$

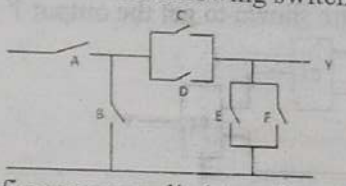
d) How many two inputs NOR gates will be used to implement two inputs NAND and XNOR gates?

e) Implement the following Boolean function with the minimum number of two inputs NAND Gate:

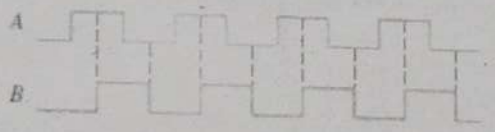
$$f(W,X,Y,Z) = \overline{(\bar{W} + \bar{X}).(\bar{Y} + \bar{Z})}$$

f) Convert $(42)_8$ in Gray code and find the BCD code for a decimal number $(478)_{10}$.

g) Find the logic expression of Y for the following switching circuit.



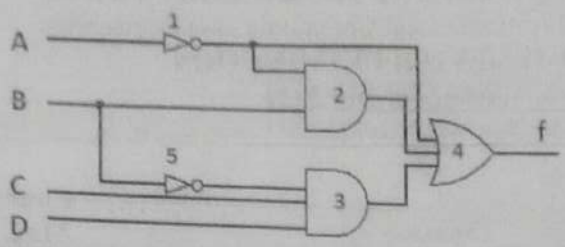
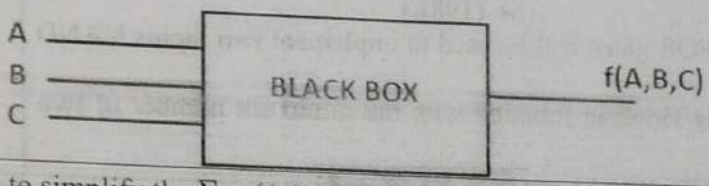
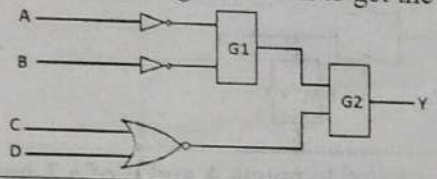
h) The waveforms in below figure are applied to points A and B of a 2-input AND gate followed by an inverter. Draw the output waveform.



i) Calculate all the minterms present in canonical SOP expression $\bar{A} + B\bar{C}$.

j) Perform the 1's and 2's complement of following numbers:
(i) $(101100)_2$ (ii) $(0.0110)_2$

2.	A Boolean function of three variables A, B, and C is defined as follows:
	f(0,0,1)=f(0,1,1)=f(1,1,0)=f(1,1,1)=1;
	f(0,0,0)=f(0,1,0)=f(1,0,0)=f(1,0,1)=0;
	Assuming complements of A, B, and C are not available. Construct the truth table and draw the logic diagram of the minimized Boolean expression. What will be the total minimum cost for realizing f using only 2-input NOR gates, 2-input AND gates, and 2-input OR gates (each having unit cost).
		5	CO1	U, R

3	Consider the below logic circuit with their gate number. Find the redundant gate. 	2.5	CO1,	A, E, C
4	Simplify the following expression by using Quine McClusky method. $F(A,B,C,D) = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + AB\bar{C}D$ Or Design the logic circuit for the Boolean function $f(A,B,C) = \sum m(0,2,3,5,7)$ using the cost effective multiplexer circuit.	5	CO2	U, A, R
5	The black box in the figure shown below consists of a minimum complexity circuit that uses only AND, OR, and NOT gates. The function f(A,B,C) = 1 whenever A and B are different and 0 otherwise. What will be the Boolean expression for the minimum complexity circuit? 	2.5	CO1, CO2	E, A, U
6	Apply K-map to simplify the $\sum m(1,7,15)$ while don't care terms are $(3,11,13)$.	2.5	CO2	R, U
7	Determine the gates G1 and G2 in the figure shown to get the output $Y = AB + \bar{C}\bar{D}$ 	2.5	CO1	R, U